

Lecture 2: Introduction to Longitudinal Data Analysis/Descriptive Statistics

January 30th, 2025

Class Outline

- Why do we use longitudinal data (brief review)
- Key features of studying change
- Person-Level (“wide”) data format pros and cons
 - Descriptive statistics
 - Profile plots
- Person-Period (“long”) data format pros and cons
 - Empirical growth plots
 - Non-parametric standardization
 - Parametric standardization
- Basic concepts

Why Do We Use Longitudinal Data?

- We want to understand how something changes over time
 - Defining feature of longitudinal studies is that measurements of the same individuals are taken repeatedly through time
 - Primary goal is to characterize the change in response over time and the factors that influence change.
- Need metrics to do this
 - Months, semesters, days, years...etc.
- We are interested in knowing the functional form of a phenomena
- With repeated measures on individuals, we can also capture within-individual change.

Review of Terminology

- Individuals/Subjects: Participants in a longitudinal study are referred to as individuals or subjects.
- Occasions: In a longitudinal study individuals are measured repeatedly at different occasions or times.
- The number of repeated observations, and their timing, can vary widely from one longitudinal study to another.
- When number and timing of the repeated measurements are the same for all individuals, study design is said to be “balanced” over time.
- Designs can be balanced, although studies may have incompleteness in data collection.

Example 1: Treatment of Lead-Exposed Children Trial

Table 1: Blood lead levels ($\mu\text{g}/\text{dL}$) at baseline, week 1, week 4, and week 6 for 8 randomly selected children.

ID	Group ^a	Baseline	Week 1	Week 4	Week 6
046	P	30.8	26.9	25.8	23.8
149	A	26.5	14.8	19.5	21.0
096	A	25.8	23.0	19.1	23.2
064	P	24.7	24.5	22.0	22.5
050	A	20.4	2.8	3.2	9.4
210	A	20.4	5.4	4.5	11.9
082	P	28.6	20.8	19.2	18.4
121	P	33.7	31.6	28.5	25.1

^a **P = Placebo; A = Succimer.**

Table 2: Mean blood lead levels (and standard deviation) at baseline, week 1, week 4, and week 6.

Group	Baseline	Week 1	Week 4	Week 6
Succimer	26.5 (5.0)	13.5 (7.7)	15.5 (7.8)	20.8 (9.2)
Placebo	26.3 (5.0)	24.7 (5.5)	24.1 (5.7)	23.2 (6.2)

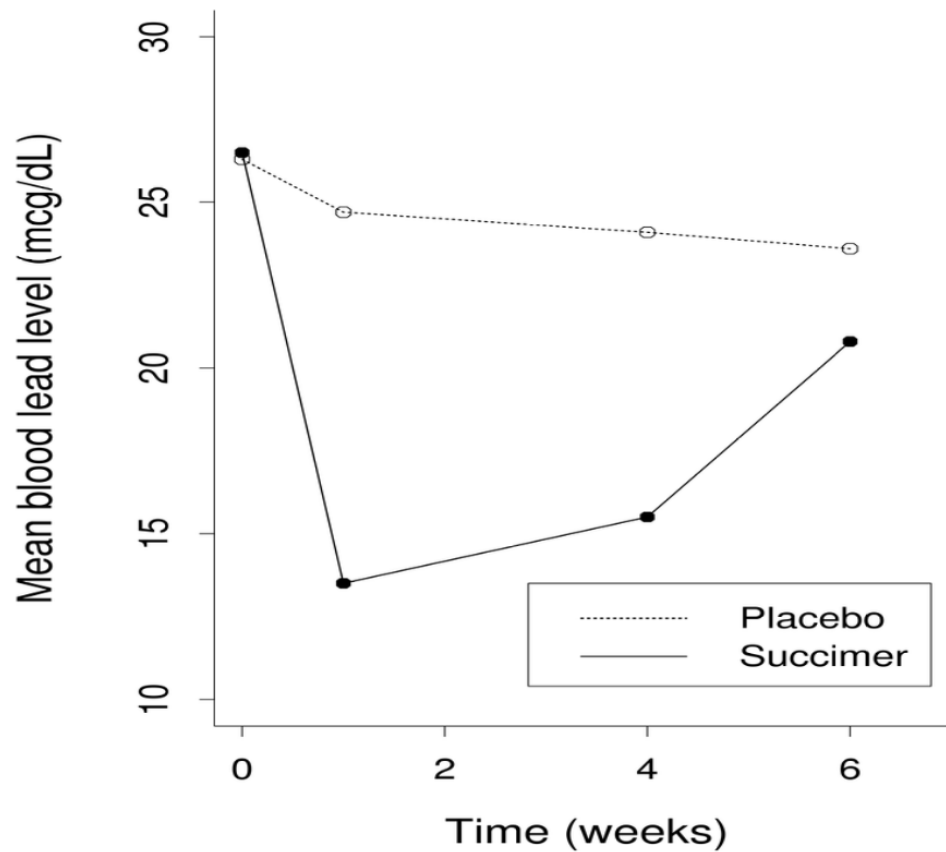


Figure 1: Plot of mean blood lead levels at baseline, week 1, week 4, and week 6 in the succimer and placebo groups.

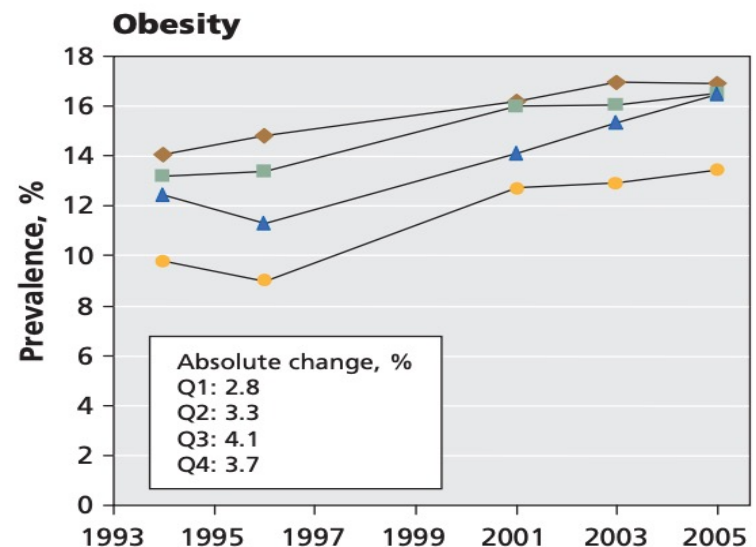
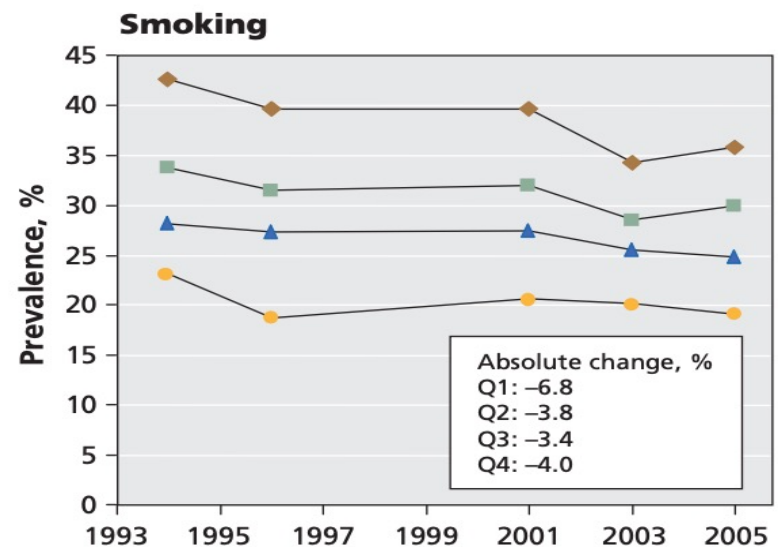
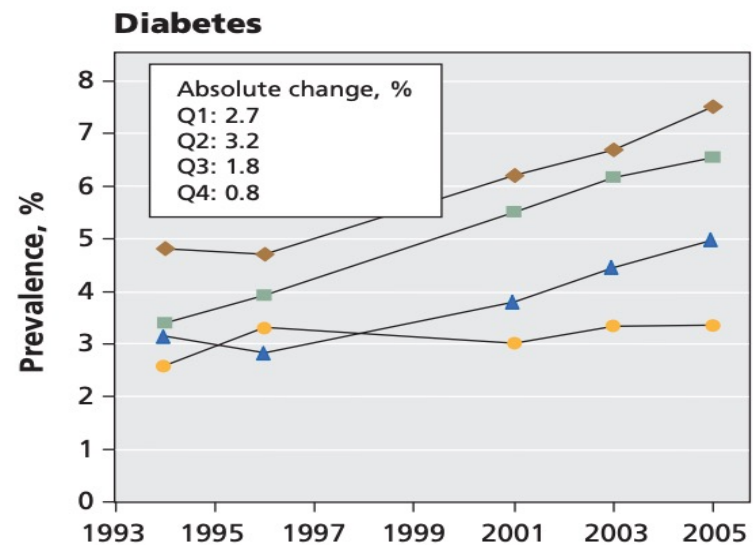
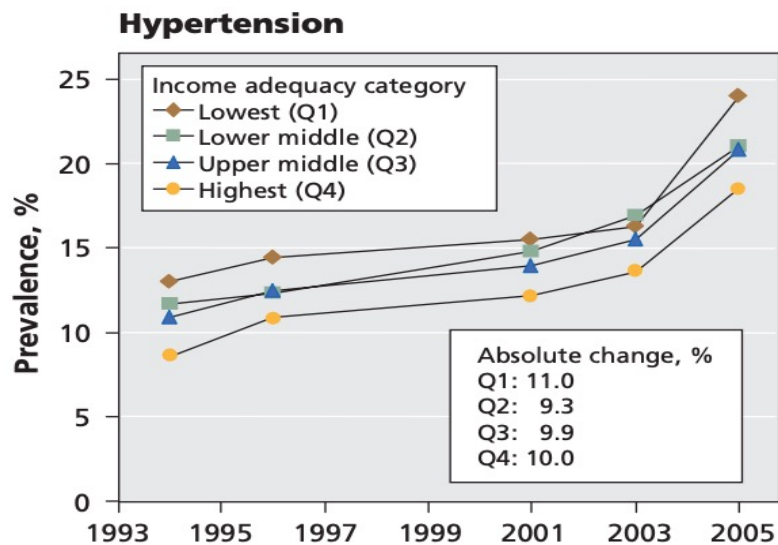
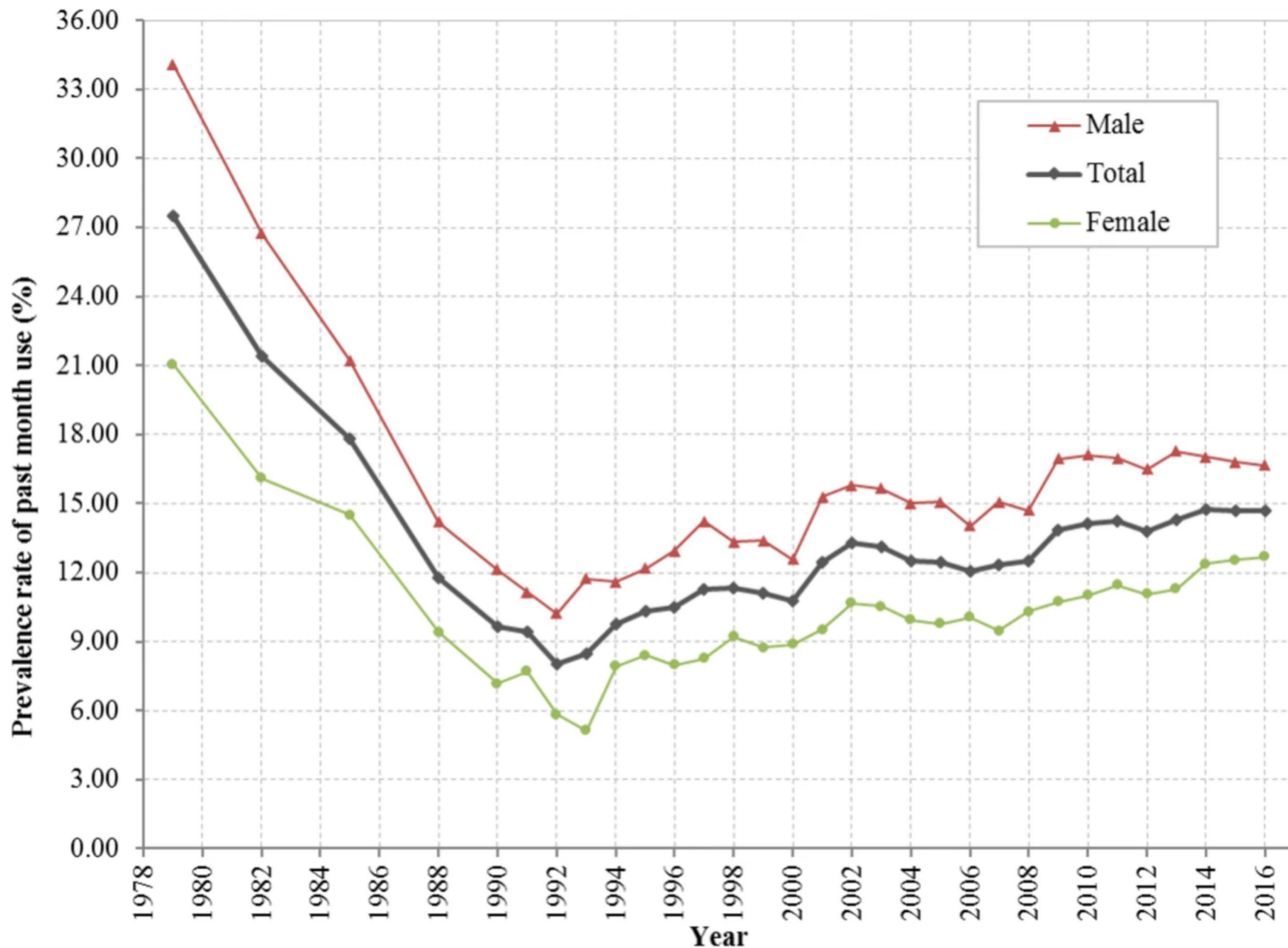


Figure 2: Age- and sex-adjusted prevalence of hypertension, diabetes, smoking and obesity from 1994 to 2005, stratified by income adequacy category.

Fig. 1

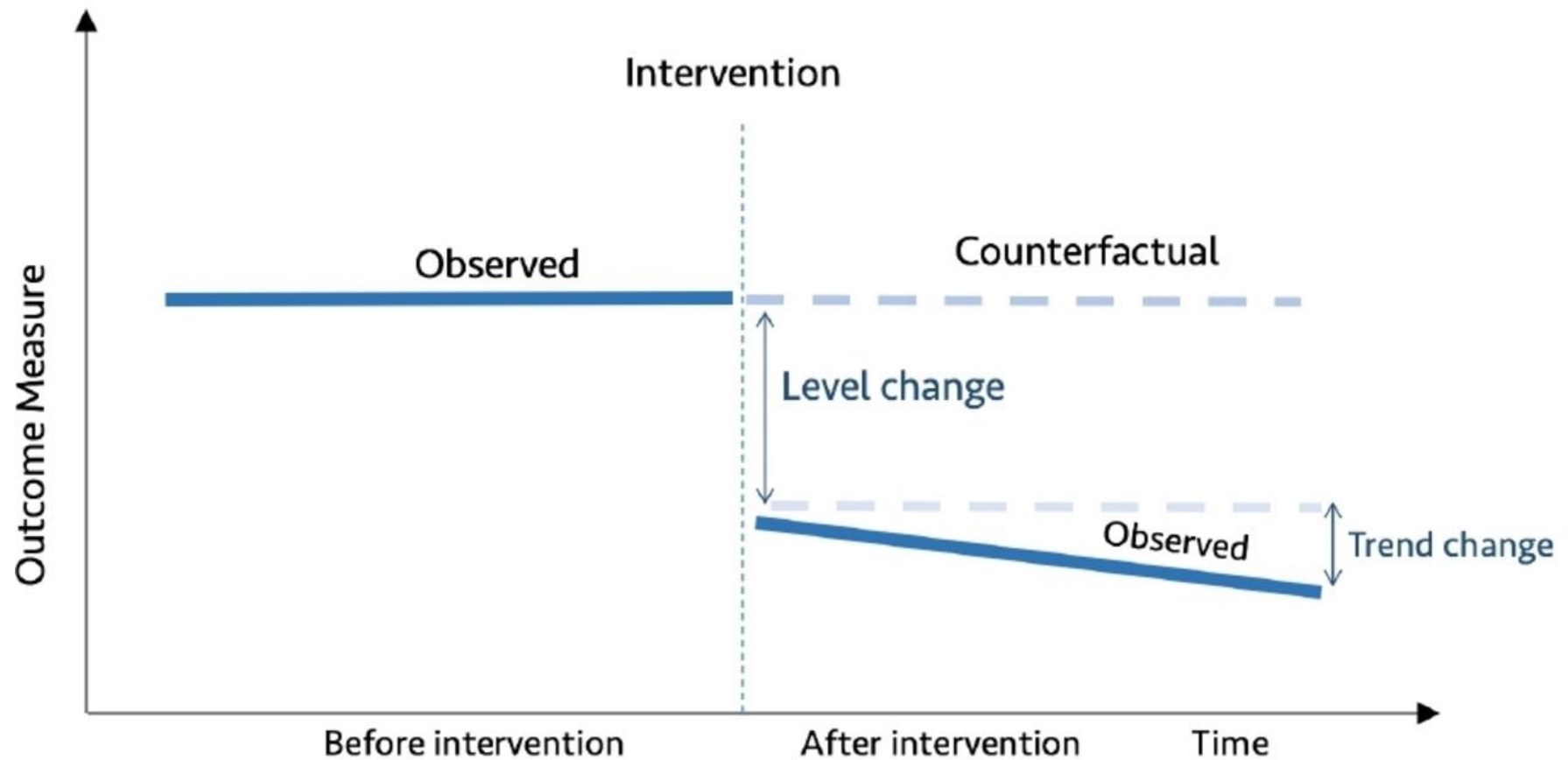
From: [Marijuana legalization and historical trends in marijuana use among US residents aged 12–25: results from the 1979–2016 National Survey on drug use and health](#)



Prevalence rate (%) of current marijuana use among US residents 12 to 25 years of age during 1979–2016, overall and stratified by gender. Derived from data from the 1979–2016 National Survey on Drug Use and Health (NSDUH)

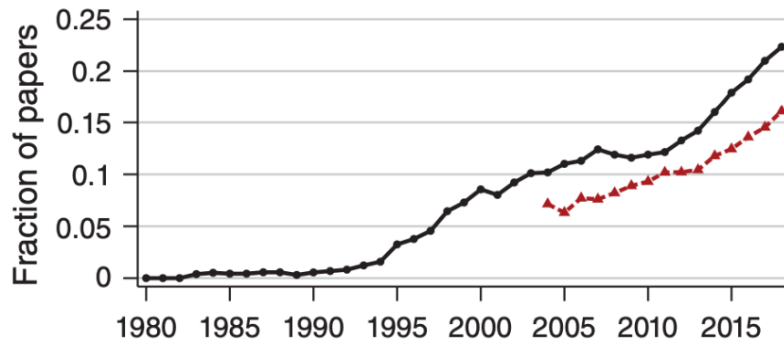
Another application: Interrupted Time Series Design

- Referred to as DID (Difference in differences) for panel data
- Method is formally discussed by Card and Krueger (1994)

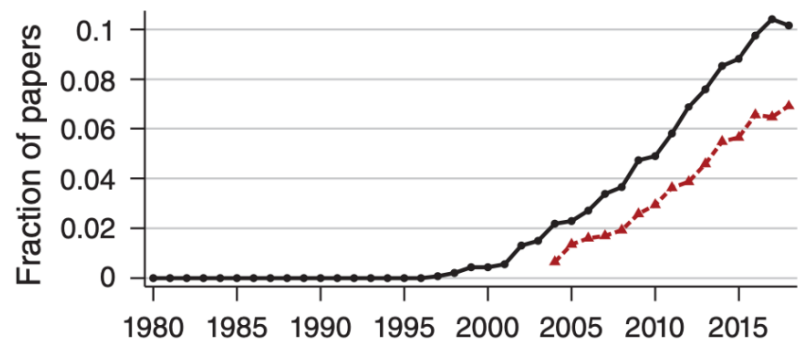


Extremely Popular in Economics

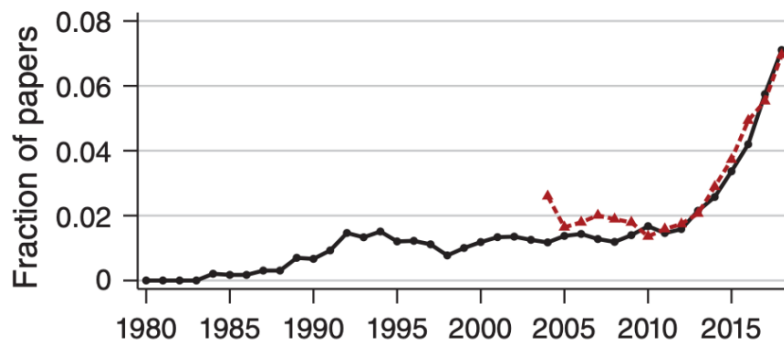
Panel A. Difference-in-differences



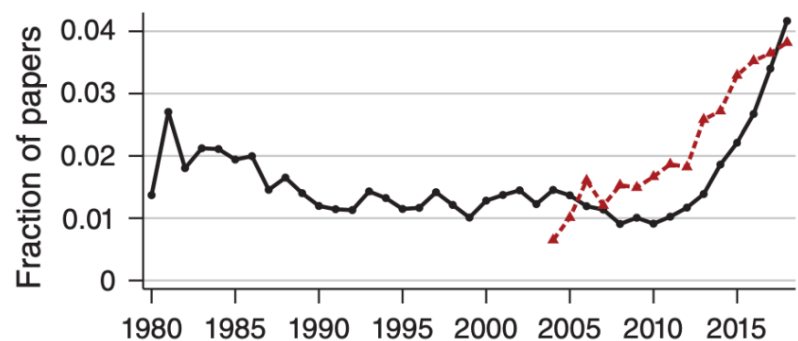
Panel B. Regression discontinuity



Panel C. Event study

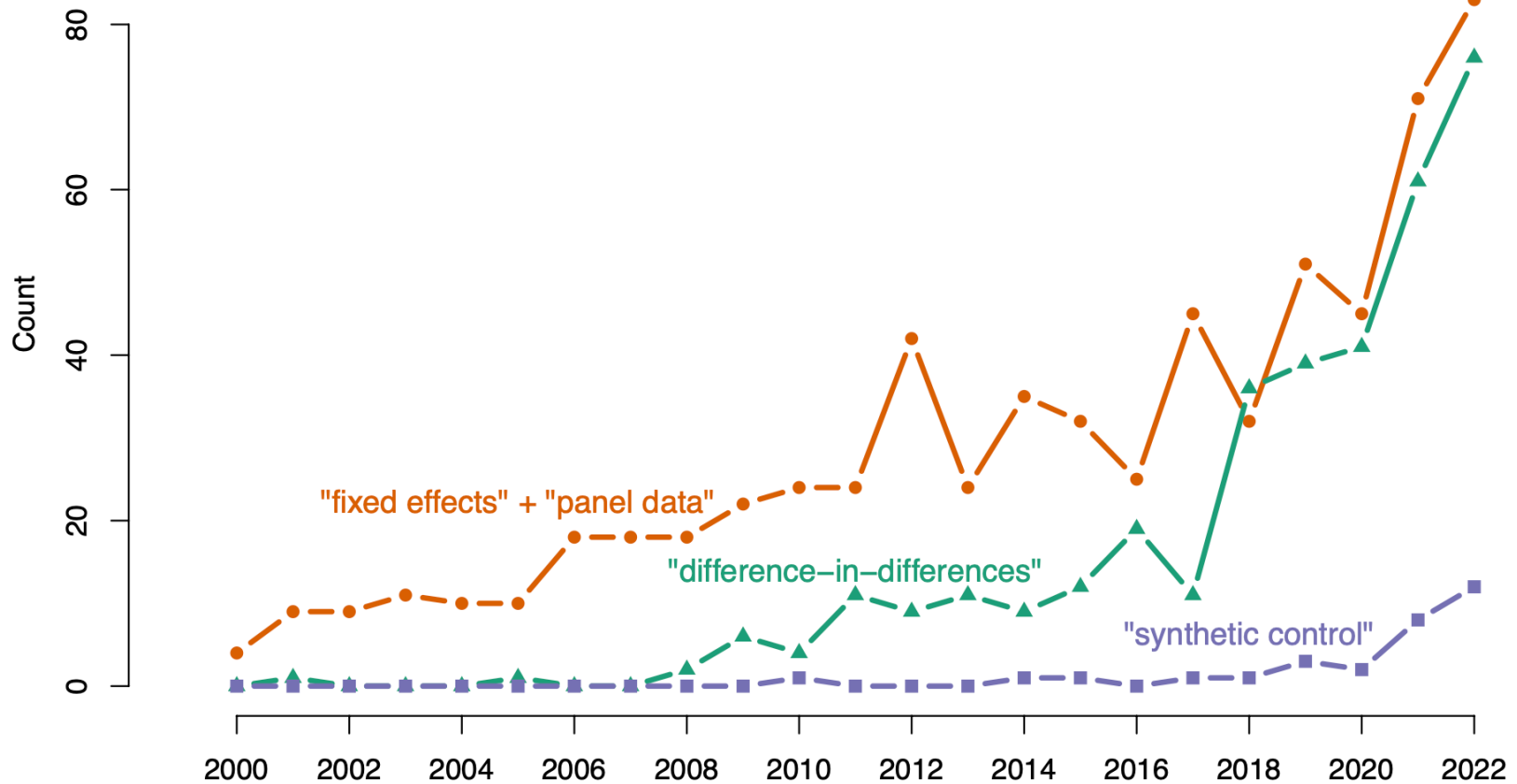


Panel D. Bunching



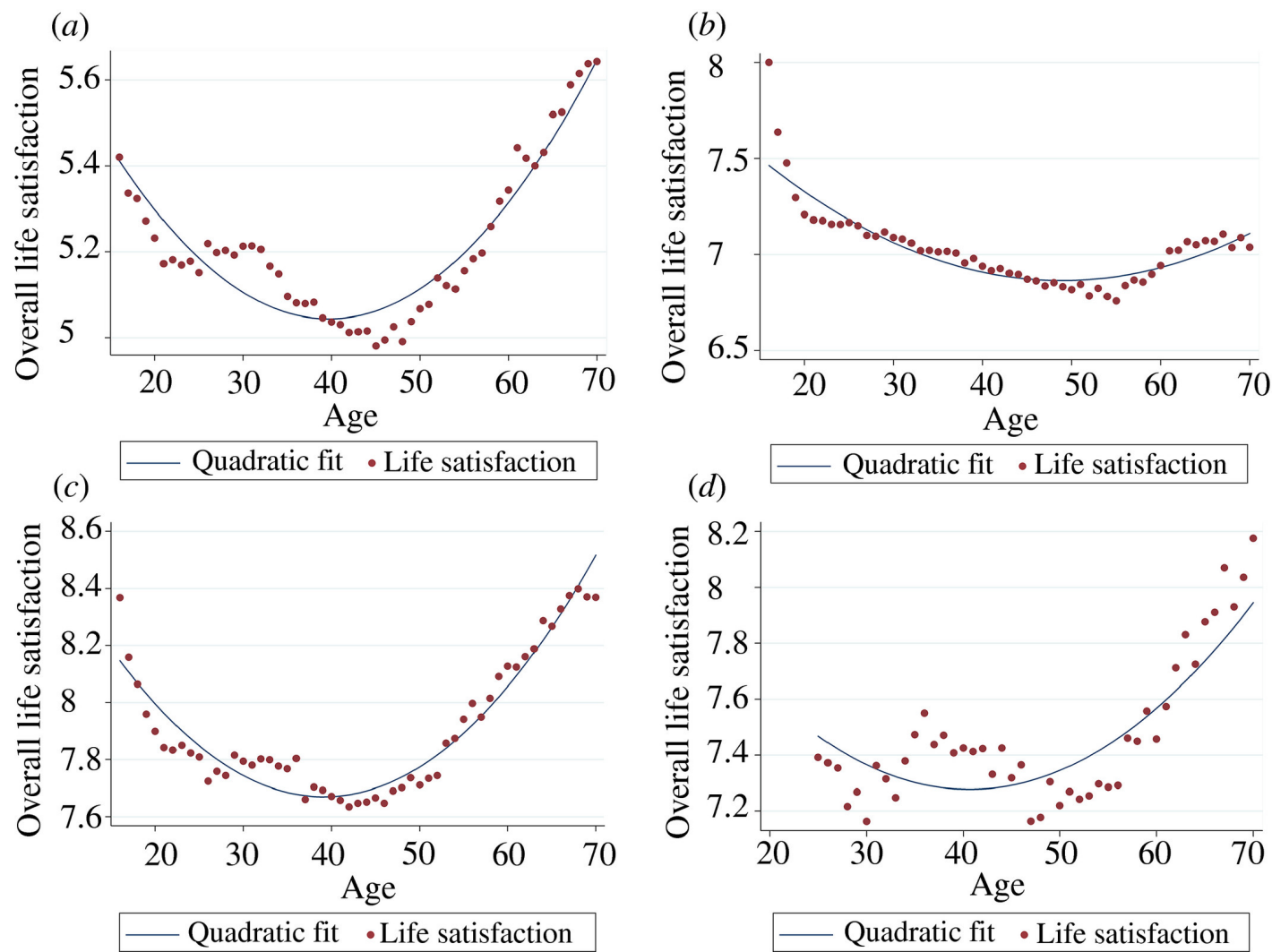
—●— NBER working papers - - -▲- - - Top-five journals

Extremely Popular in Politics



Longitudinal Evidence for a Midlife Nadir in Well-being: Results from Four Data Sets

Human



Key features of studying change

(Singer and Willet page 9-15)

- Three or more time points
 - The problem with serial cross-sectional data
 - Bias due to differences in history, cohort, etc.
 - Choosing a metric for time
 - Theory?
 - What makes conceptual sense?
 - What metric of time makes most sense for your question?
 - An outcome whose value changes systematically over time
 - Select outcome wisely, make sure valid at each time point and that you are measuring the same thing.

- The outcome variable can take on different distributional properties
 - Continuous (e.g., Viral Load)
 - Binary (e.g., substance use/no substance use)
 - Count (e.g., number seizures in the last 30 days)
- Attrition
 - Participants or units can have missing data

More Features: Correlation

- An aspect of longitudinal data that complicates their statistical analysis is that repeated measures on the same individual are usually positively correlated.
- This violates the fundamental assumption of independence that is the cornerstone of many statistical techniques.
- Why are longitudinal data correlated?
- What are the potential consequences of not accounting for correlation among longitudinal data in the analysis?

More Features: Heteroscedasticity

- Another aspect of longitudinal data that complicates their statistical analysis is heterogeneous variability.
- That is, the variability of the outcome at the end of the study is often discernibly different than the variability at the start of the study.
- This violates the assumption of homoscedasticity that is the basis for standard linear regression techniques.
- Two difficulties of longitudinal data that complicate their statistical analysis: (i) repeated measures on the same individual are usually positively correlated, and (ii) variability is often heterogeneous across measurement occasions.

Wide and Long Form

Person-Level (“wide”) data format

(Singer and Willet page 17-22)

. list

- In this format each person has one row of data consisting of multiple variables
- There is no time variable
- The repeated variables are listed as an individual column of data (e.g. tol11-tol15)

	id	tol11	tol12	tol13	tol14	tol15	male	exposure
1.	9	2.23	1.79	1.9	2.12	2.66	0	1.54
2.	45	1.12	1.45	1.45	1.45	1.99	1	1.16
3.	268	1.45	1.34	1.99	1.79	1.34	1	.9
4.	314	1.22	1.22	1.55	1.12	1.12	0	.81
5.	442	1.45	1.99	1.45	1.67	1.9	0	1.13
6.	514	1.34	1.67	2.23	2.12	2.44	1	.9
7.	569	1.79	1.9	1.9	1.99	1.99	0	1.99
8.	624	1.12	1.12	1.22	1.12	1.22	1	.98
9.	723	1.22	1.34	1.12	1	1.12	0	.81
10.	918	1	1	1.22	1.99	1.22	0	1.21
11.	949	1.99	1.55	1.12	1.45	1.55	1	.93
12.	978	1.22	1.34	2.12	3.46	3.32	1	1.59
13.	1105	1.34	1.9	1.99	1.9	2.12	1	1.38
14.	1542	1.22	1.22	1.99	1.79	2.12	0	1.44
15.	1552	1	1.12	2.23	1.55	1.55	0	1.04
16.	1653	1.11	1.11	1.34	1.55	2.12	0	1.25

NOTE: if you would like a subset of the data use the stata *syntax list in 1/10* (this will give you only the first 10 cases)

Person-Level (“wide”) data format

(Singer and Willet page 17-22)

. list in 1/7

- Pros-

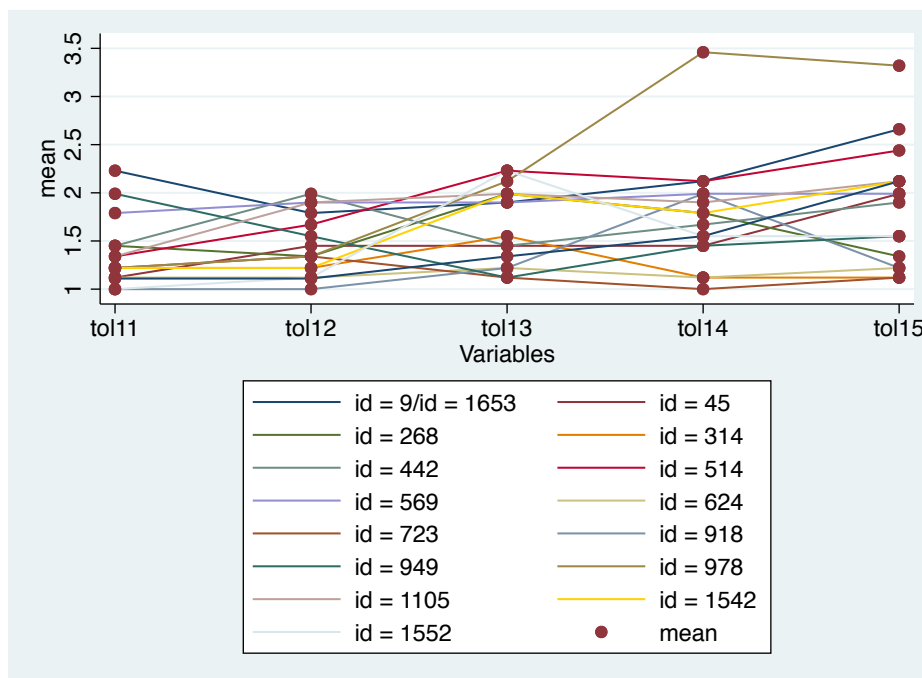
- Easy to examine general trends in data
- You can “eyeball” the functional form of the data
- Great for data management

	id	tol11	tol12	tol13	tol14	tol15	male	exposure
1.	9	2.23	1.79	1.9	2.12	2.66	0	1.54
2.	45	1.12	1.45	1.45	1.45	1.99	1	1.16
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6.	514	1.34	1.67	2.23	2.12	2.44	1	.9
7.	569	1.79	1.9	1.9	1.99	1.99	0	1.99

- Cons-

- Non-informative summaries
- No time variable
- Inefficient if there is unequal spacing
- Cannot really handle time-varying predictors

. profileplot tol11 tol12 tol13 tol14 tol15, by(id)



NOTE: Must download and run the profileplot adofile to use this option-
(<https://stats.oarc.ucla.edu/stata/ado/analysis/>)

Person-Level (“wide”) data format

(Singer and Willet page 17-22)

. list

- Pros-
 - Easy to examine general trends in data
 - You can “eyeball” the functional form of the data
 - Great for data management
- Cons-
 - Non-informative summaries
 - No time variable
 - Inefficient if there is unequal spacing
 - Cannot really handle time-varying predictors

	id	tol11	tol12	tol13	tol14	tol15	male	exposure
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2.	45	1.12	1.45	1.45	1.45	1.99	1	1.16
3.	268	1.45	1.34	1.99	1.79	1.34	1	.9
4.	314	1.22	1.22	1.55	1.12	1.12	0	.81
5.	442	1.45	1.99	1.45	1.67	1.9	0	1.13
6.	514	1.34	1.67	2.23	2.12	2.44	1	.9
7.	569	1.79	1.9	1.9	1.99	1.99	0	1.99
8.	624	1.12	1.12	1.22	1.12	1.22	1	.98
9.	723	1.22	1.34	1.12	1	1.12	0	.81
10.	918	1	1	1.22	1.99	1.22	0	1.21
11.	949	1.99	1.55	1.12	1.45	1.55	1	.93
12.	978	1.22	1.34	2.12	3.46	3.32	1	1.59
13.	1105	1.34	1.9	1.99	1.9	2.12	1	1.38
14.	1542	1.22	1.22	1.99	1.79	2.12	0	1.44
15.	1552	1	1.12	2.23	1.55	1.55	0	1.04
16.	1653	1.11	1.11	1.34	1.55	2.12	0	1.25

. pwcorr

	tol11	tol12	tol13	tol14	tol15	sig
tol11	1.0000					
tol12	0.6573 0.0057	1.0000				
tol13	0.0619 0.8197	0.2476 0.3553	1.0000			
tol14	0.1408 0.6031	0.2056 0.4449	0.5872 0.0168	1.0000		
tol15	0.2635 0.3240	0.3923 0.1329	0.5692 0.0214	0.8255 0.0001	1.0000	

Person-Period ("long") data format

(Singer and Willet page 22-32)

- Each person has a record for each measurement occasion
- There is a time variable
- Time varying and time-invariant predictors
- The id variable is repeated over the different time (e.g. 5 time points for tolerance means id is repeated 5 times).

```
. reshape long tol, i(id) j(age)
(note: j = 11 12 13 14 15)
```

Data	wide	->	long
Number of obs.	16	->	80
Number of variables	8	->	5
j variable (5 values)		->	age
xij variables:			
	tol11 tol12 ... tol15	->	tol

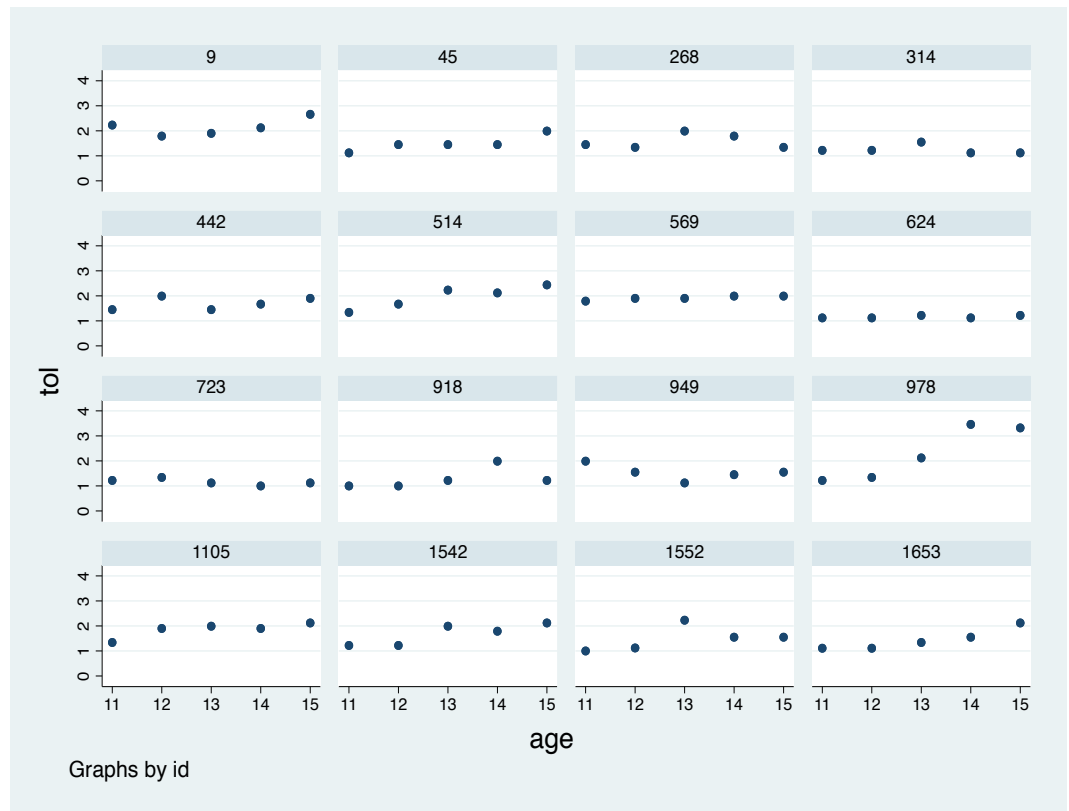
```
. list in 1/16
```

	id	age	tol	male	exposure
1.	9	11	2.23	0	1.54
2.	9	12	1.79	0	1.54
3.	9	13	1.9	0	1.54
4.	9	14	2.12	0	1.54
5.	9	15	2.66	0	1.54
6.	45	11	1.12	1	1.16
7.	45	12	1.45	1	1.16
8.	45	13	1.45	1	1.16
9.	45	14	1.45	1	1.16
10.	45	15	1.99	1	1.16
11.	268	11	1.45	1	.9
12.	268	12	1.34	1	.9
13.	268	13	1.99	1	.9
14.	268	14	1.79	1	.9
15.	268	15	1.34	1	.9
16.	314	11	1.22	0	.81

Person-Period ("long") data format

(Singer and Willet page 25)

```
. graph twoway scatter tol age, by(id) ylabel(0(1)4) xlabel(11(1)15)
```



- Descriptive analyses

- How do people change over time?
- Is there growth?
- What is the functional form?

- Empirical growth plot

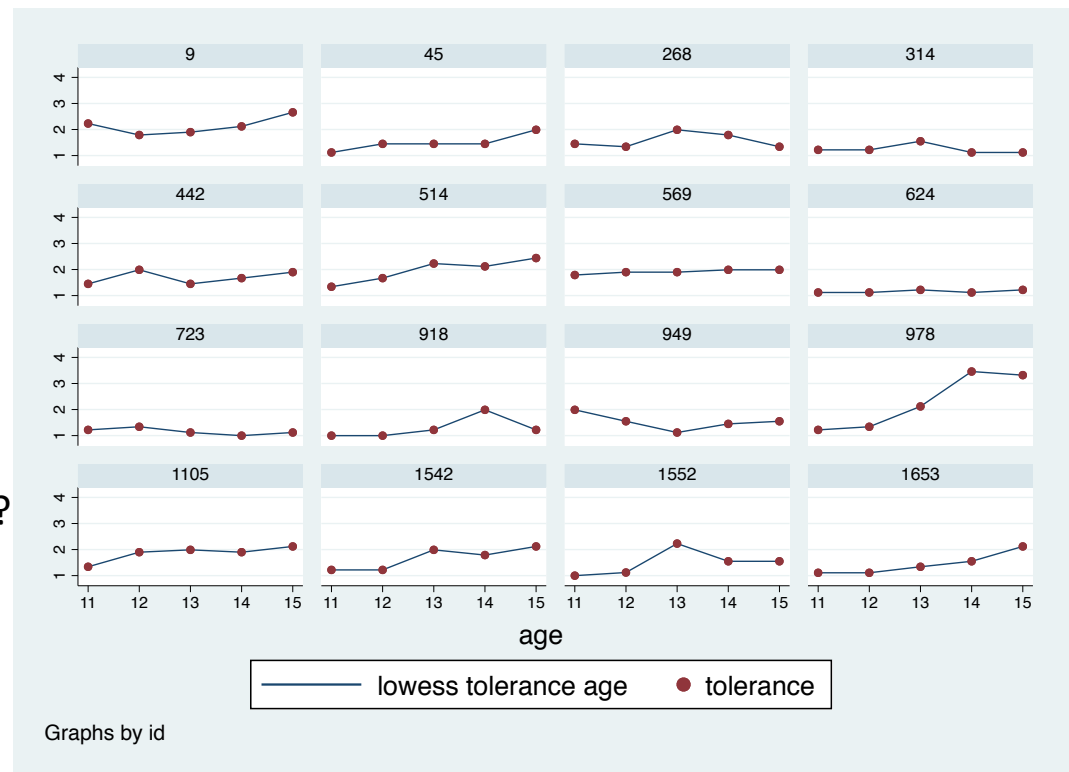
- Who is increasing, who is decreasing, who is increasing the most?
- What information can you garner from this plot?

Person-Period ("long") data format

(Singer and Willet pages 26 and 27)

- Non-parametric standardized approach
 - No functional form
 - “Let the data speak for themselves”
- What do you notice about the data?
 - Is the trajectory a bit clearer now?

. graph twoway (lowess tolerance age)(scatter tolerance age), by(id)

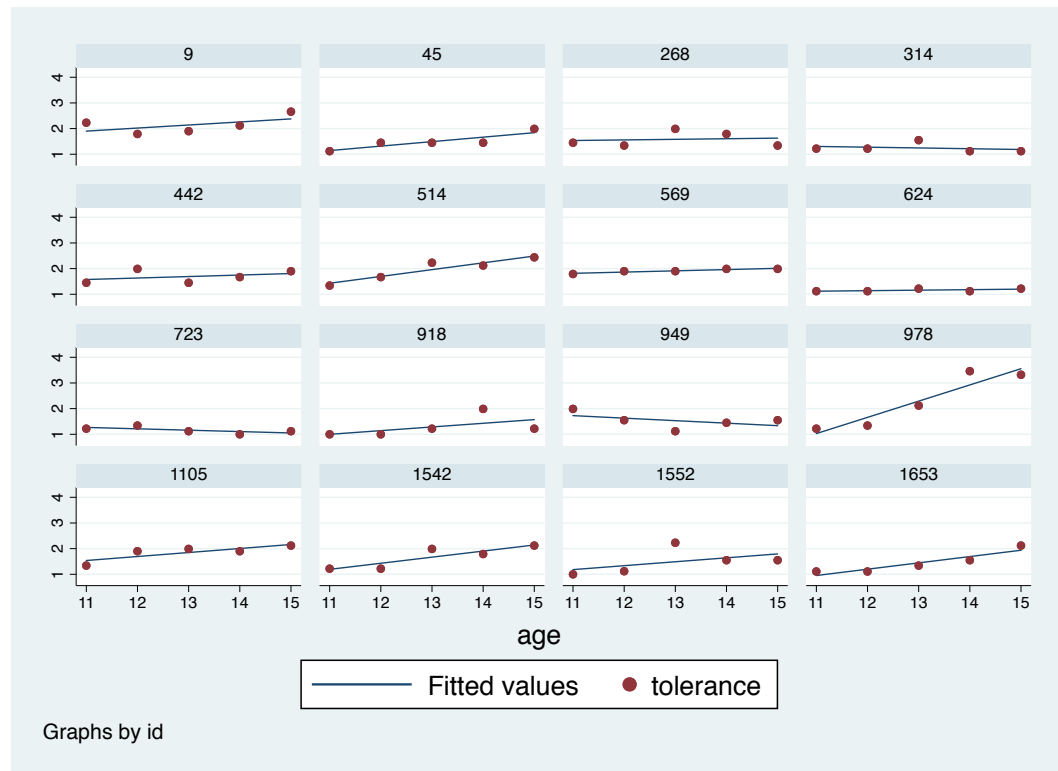


Person-Period ("long") data format

(Singer and Willet pages 32-34)

- Parametric standardized approach
 - OLS linear regression
 - Fitted OLS trajectories superimposed on empirical growth plots for participants
- What do you notice about the data?
 - Anything different?
 - Confirmatory?

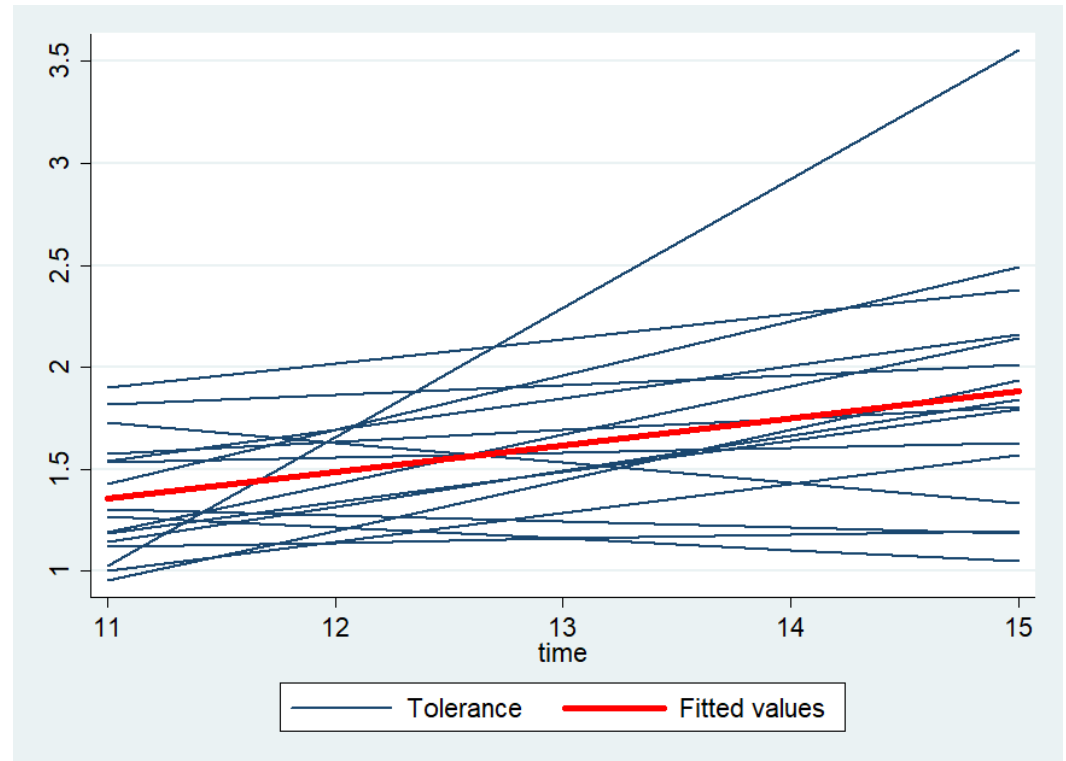
. graph twoway (lfit tolerance age)(scatter tolerance age), by(id)



Person-Period ("long") data format

(see dofile in Lecture 2 folder, Singer and Willet pages 34 and 38)

- Parametric standardized approach
 - OLS linear regression
 - Fitted OLS trajectories superimposed on empirical growth plots for participants
 - **Overall mean trajectory line**
- What do you notice about the data?
 - Anything different?
 - Confirmatory?



Notes

- Average change trajectory (page 34)
 - Confirmatory for functional form
- Will look at model fit and change due to a time invariant covariate starting next class period.

Cortisol Example

Blunted Diurnal Cortisol Activity in Healthy Adults with Childhood Adversity

Yuliya I. Kuras, Naomi Assaf, Myriam V. Thoma, Danielle Gianferante, Luke Hanlin¹, Xuejie Chen, Alexander Fiksdal and Nicolas Rohleder

Question: Is childhood adversity associated with a blunted cortisol response in adulthood?

Hypothesis: we hypothesized that adults with low-to-moderate childhood adversity would have altered hypothalamic-pituitary-adrenal (HPA) axis functioning, as evidenced by a blunted diurnal cortisol slope and altered cortisol awakening response (CAR).

Cortisol Example

Blunted Diurnal Cortisol Activity in Healthy Adults with Childhood Adversity

Yuliya I. Kuras, Naomi Assaf, Myriam V. Thoma, Danielle Gianferante, Luke Hanlin¹, Xuejie Chen, Alexander Fiksdal and Nicolas Rohleder

Details:

- Age 18 - 65
- 60 participants (31 males; 30 females)
- Trauma questionnaire
- Cortisol collection – wake-up, 30mins, 1h, 4h, 9h, 13h for 2 days

Cortisol Example

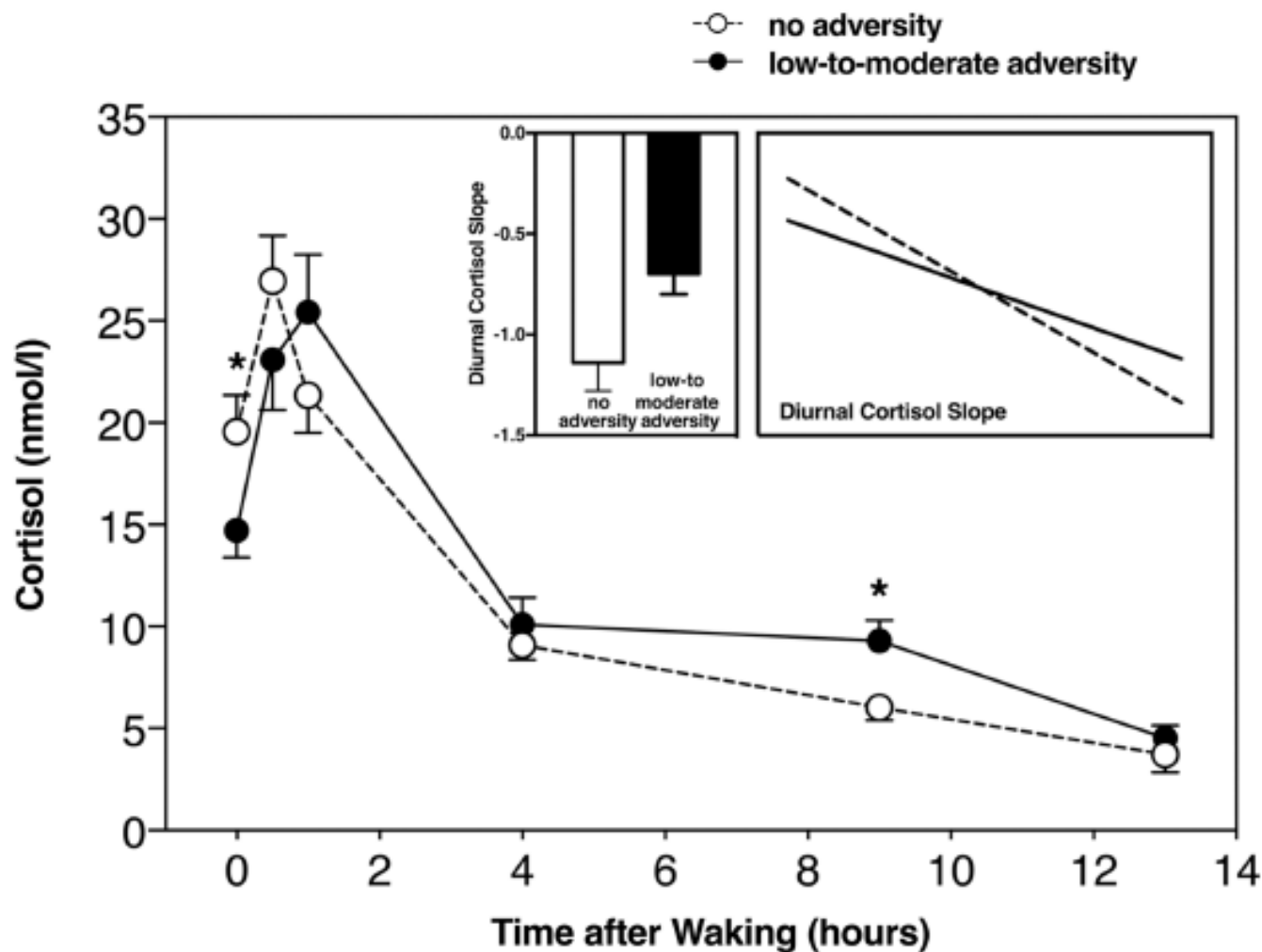
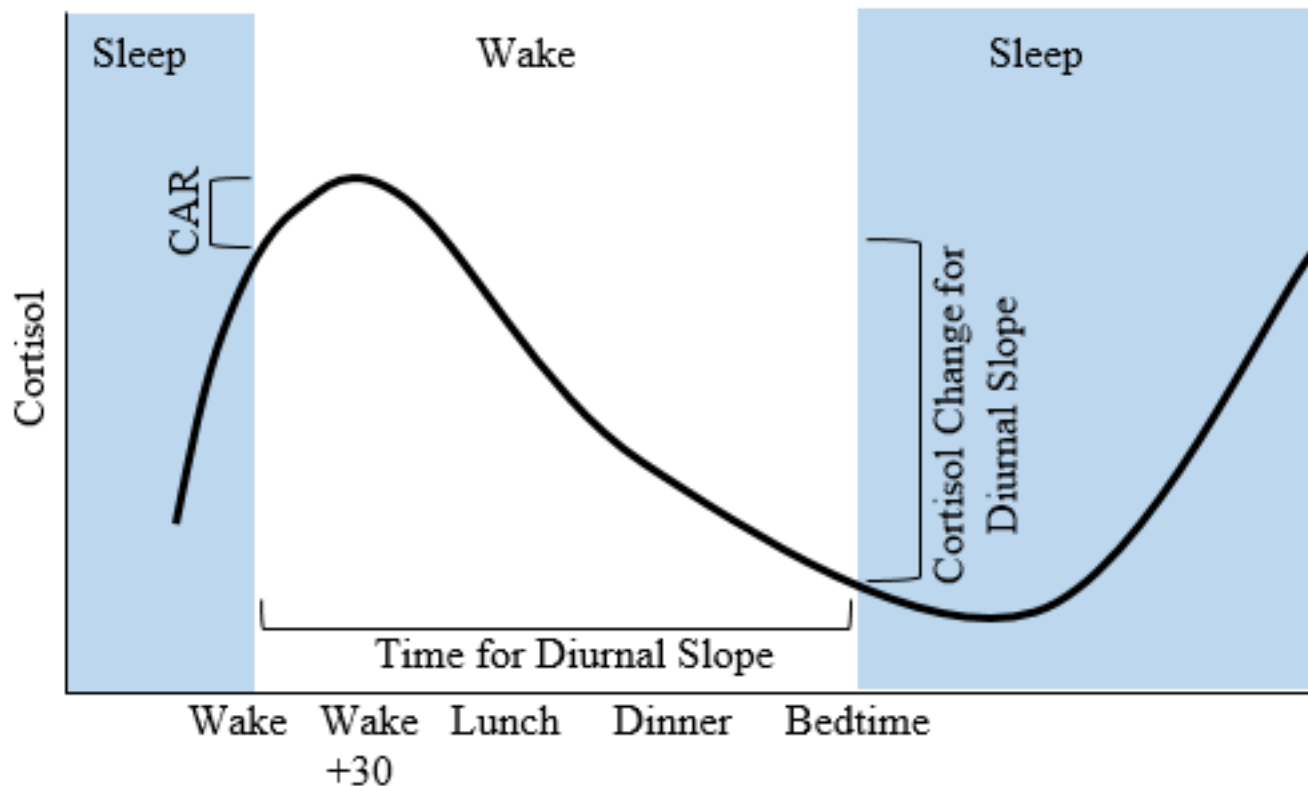


FIGURE 2 | Diurnal cortisol slope including all time-points in those with and without childhood adversity, with slope score difference between the two groups (left insert), and diurnal slope without CAR (right insert). *Significant at $p < 0.05$.

Notes on Cortisol Exercise

- Remember...
 - Choosing a metric for time is based on concept or theory that changes systematically overtime



Notes on Cortisol Exercise

- Based on knowing that cortisol has a diurnal pattern, what would you be looking for in your data?
 1. If the linear pattern (e.g., decreasing or increasing trend) changes over time?
 2. What are the factors that predict changes in the intercept and linear change over time?
 3. What are the factors that predict changes in quadratic pattern (e.g. flatter curve, steeper curve) over time?

Formal Description of Data Format

- Let Y_{ij} denote the response variable for the i th individual ($i = 1, \dots, N$) at the j th occasion ($j = 1, \dots, n$).
- If the repeated measures are assumed to be equally-separated in time, this notation will be sufficient.
 - You can refine it to handle the case where repeated measures are unequally-separated and unbalanced over time.
- We can represent the n observations on the N individuals in a two-dimensional array, with rows corresponding to individuals and columns corresponding to the responses at each occasion.

Table 4: Tabular representation of longitudinal data, with n repeated observations on N individuals.

Individual	Occasion				
	1	2	3	...	n
1	y_{11}	y_{12}	y_{13}	...	y_{1n}
2	y_{21}	y_{22}	y_{23}	...	y_{2n}
.	.	.	.	...	.
.	.	.	.	...	.
.	.	.	.	...	.
N	y_{N1}	y_{N2}	y_{N3}	...	y_{Nn}

Vector Notation

- We can group the n repeated measures on the same individual into a $n \times 1$ response vector:

$$Y_i = \begin{pmatrix} Y_{i1} \\ Y_{i2} \\ \vdots \\ Y_{in} \end{pmatrix}$$

- Alternatively, we can denote the response vectors Y_i as

$$Y_i = (Y_{i1}, Y_{i2}, \dots, Y_{in})'.$$

Review

- Denote the expectation or mean of Y_{ij} by $\mu_j = E(Y_{ij})$, where $E(\cdot)$ can be thought of as a long-run average (over individuals).
 - The mean, μ_j , provides a measure of the location of the center of the distribution of Y_{ij} .
- The variance provides a measure of the spread or dispersion of the values of Y_{ij} around its respective mean:

$$\sigma_j^2 = E[Y_{ij} - E(Y_{ij})]^2 = E(Y_{ij} - \mu_j)^2.$$

- The positive square-root of the variance, σ_j , is known as the standard deviation.
- The covariance between two variables, say Y_{ij} and Y_{ik} is

$$\sigma_{jk} = E[(Y_{ij} - \mu_j)(Y_{ik} - \mu_k)],$$

is a measure of the linear dependence between Y_{ij} and Y_{ik}

- When the covariance is zero, there is no linear dependence between the responses at the two occasions.

- The correlation between Y_{ij} and Y_{ik} is denoted by

$$\rho_{jk} = \frac{E[(Y_{ij} - \mu_j)(Y_{ik} - \mu_k)]}{\sigma_j \sigma_k},$$

where σ_j and σ_k are the standard deviations of Y_{ij} and Y_{ik} .

- The correlation, unlike covariance, is a measure of dependence free of scales of measurement of Y_{ij} and Y_{ik} .
- By definition, correlation must take values between -1 and 1 .
 - A correlation of 1 or -1 is obtained when there is a perfect linear relationship between the two variables.

- For the vector of repeated measures,

$$Y_i = (Y_{i1}, Y_{i2}, \dots, Y_{in})'.$$

We define the variance-covariance matrix, $Cov(Y_i)$,

$$\begin{aligned} Cov \begin{pmatrix} Y_{i1} \\ Y_{i2} \\ \vdots \\ Y_{in} \end{pmatrix} &= \begin{pmatrix} Var(Y_{i1}) & Cov(Y_{i1}, Y_{i2}) & \cdots & Cov(Y_{i1}, Y_{in}) \\ Cov(Y_{i2}, Y_{i1}) & Var(Y_{i2}) & \cdots & Cov(Y_{i2}, Y_{in}) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(Y_{in}, Y_{i1}) & Cov(Y_{in}, Y_{i2}) & \cdots & Var(Y_{in}) \end{pmatrix} \\ &= \begin{pmatrix} \sigma_1^2 & \sigma_{12} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \cdots & \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \cdots & \sigma_n^2 \end{pmatrix}, \end{aligned}$$

where $Cov(Y_{ij}, Y_{ik}) = \sigma_{jk} = \sigma_{kj} = Cov(Y_{ik}, Y_{ij})$.

- We can also define the correlation matrix, $Corr(Y_i)$,

$$Corr(Y_i) = \begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1n} \\ \rho_{21} & 1 & \cdots & \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{n1} & \rho_{n2} & \cdots & 1 \end{pmatrix}.$$

- This matrix is symmetric in the sense that $Corr(Y_{ij}, Y_{ik}) = \rho_{jk} = \rho_{kj} = Corr(Y_{ik}, Y_{ij})$.

Example 1: Treatment of Lead-Exposed Children Trial

- We restrict attention to the data from placebo group.
- Data consist of 4 repeated measurements of blood lead levels obtained at baseline (or week 0), weeks 1, 4, and 6.
- The inter-dependence (or time-dependence) among the four repeated measures of blood lead level can be examined by constructing a scatter-plot of each pair of repeated measures.
- Examination of the correlations confirms that they are all positive and tend to decrease with increasing time separation.

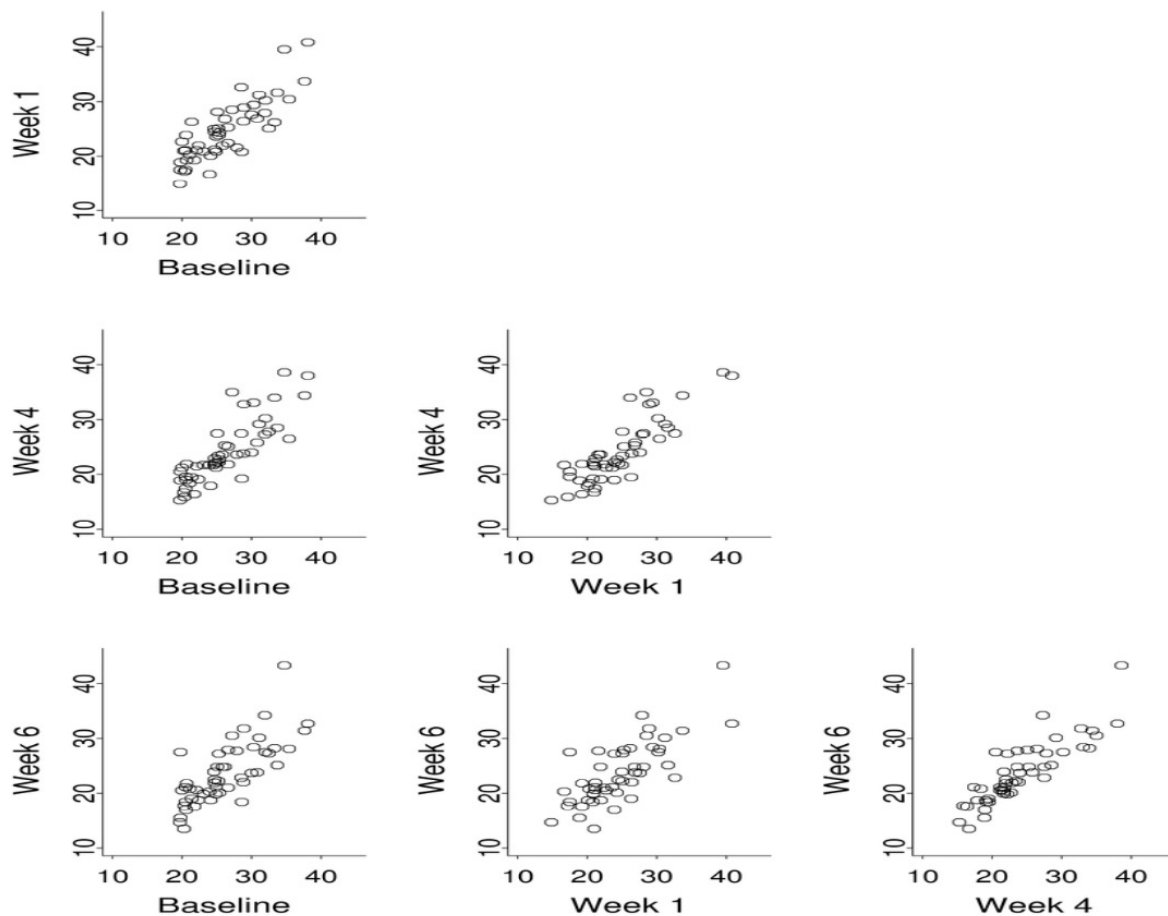


Figure 6: Pairwise scatter-plots of blood lead levels at baseline, week 1, week 4, and week 6 for children in the placebo group.

Table 5: Estimated covariance matrix for the blood lead levels at baseline, week 1, week 4, and week 6 for children in the placebo group of the TLC trial.

Covariance Matrix			
25.2	22.8	24.2	18.4
22.8	29.8	27.0	20.5
24.2	27.0	33.0	26.6
18.4	20.5	26.6	38.7

Table 6: Estimated correlation matrix for the blood lead levels at baseline, week 1, week 4, and week 6 for children in the placebo group of the TLC trial.

Correlation Matrix			
1.00	0.83	0.84	0.59
0.83	1.00	0.86	0.60
0.84	0.86	1.00	0.74
0.59	0.60	0.74	1.00

Empirical observations about the nature of the correlation among repeated measures in longitudinal studies:

1. correlations are positive,
2. correlations decrease with increasing time separation,
3. correlations between repeated measures rarely ever approach zero, and
4. correlation between a pair of repeated measures taken very closely together in time rarely approaches one.

What If Ignoring Correlation

- Potential impact of ignoring correlation can be illustrated using data from the Treatment of Lead-Exposed Children Trial.
- For simplicity, consider only the first two repeated measures, taken at week 0 and week 1.
- It is of interest to determine the change in the mean response over time.
- An estimate of change is given by

$$\hat{\delta} = \hat{\mu}_2 - \hat{\mu}_1,$$

where $\hat{\mu}_j = ?$

- In the TLC trial, the estimate of change in the succimer group is -13.0 (or $13.5 - 26.5$).
- For inferences, we also need a standard error, i.e. variance of $\hat{\delta}$

$$\text{Var}(\hat{\delta}) = \text{Var} \left\{ \frac{1}{N} \sum_{i=1}^N (Y_{i2} - Y_{i1}) \right\} = \frac{1}{N} (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}) .$$

- Last term, $-2\sigma_{12}$, accounts for the correlation among the repeated measures.
- Substituting estimates of the variances and covariance into this expression:

$$\widehat{\text{Var}}(\hat{\delta}) = \frac{1}{50} \{25.2 + 58.9 - 2(15.5)\} = 1.06.$$

- What if we had ignored that the data are correlated and proceeded with an analysis assuming all observations are independent?
- Independence $\Rightarrow$ zero covariance.
- Leading to (incorrect) estimate of the variance of $\hat{\delta}$

$$\frac{1}{50}(25.2 + 58.9) = 1.68,$$

which is approximately 1.6 times larger.

- Here, ignoring the correlation results in:
 1. standard errors that are too large
 2. confidence intervals that are too wide
 3. p-values for the test of $H_0: \delta = 0$ that are too large
- In general, failure to take account of the correlation (covariance) among the repeated measures will result in incorrect estimates of the sampling variability and can lead to quite misleading scientific inferences.

Summary

- Primary goal of a longitudinal study is to characterize the change in response over time and the factors that influence change.
- Know thy data
 - Start with simply looking at the data for key indicators of data quality and form
 - Missingness
 - Autocorrelation
 - Functional form
- Once you have an idea of what your outcome looks like, you will then dig a little deeper and decide, preliminarily, on a functional form
- Longitudinal data require somewhat more sophisticated statistical techniques because: (i) repeated measures on the same individual are usually positively correlated, and (ii) variability is often heterogeneous across measurement occasions.
- Correlation and heterogeneous variability must be accounted for in order to obtain valid inferences about change in response over time.

Next Class

- What are the key takeaways from descriptive analyses
- Time invariant predictors and descriptive analyses
- Why do we use multi-level models of change?
- Level-1 individual model of change (individual growth model)

Class Exercise 1

- Use the following link and your text book to recreate (National Youth Survey – Tolerance of antisocial behavior)

Stata

<https://stats.idre.ucla.edu/stata/examples/alda/chapter2/applied-longitudinal-data-analysis-modeling-change-and-event-occurrenceby-judith-d-singer-and-john-b-willett-chapter-2-exploring-longitudinal-data-on/>

R

<https://stats.oarc.ucla.edu/r/examples/alda/r-applied-longitudinal-data-analysis-ch-2/>

Class Exercise 2

- Flint Adolescent Study
- Depression at year 1-5 (15-19 years of age)
- Sex (male; female)
- Repeat same exercise as with the “tolerance” example.